

SC1006
Computer Organization and Architecture

Tutorial 5

#5.1: Input/Output Interface

Table 1: Processor Information

Overview	Peripheral	Electrical Interface
• Von-Neumann Architecture • 12 Data Registers (32-bit) • 256 Kbyte on-chip SRAM, 512 Kbyte on-chip Flash, • 2 Kbyte on-chip EEPROM • 4 Kbyte Cache (Data and Instruction) • 400 Mhz CPU clock • 200 Mhz External Parallel Bus interface (16-bits) • DMA Controller • Memory Management Unit	• SPI Max clock speed: 50 Mhz • UART Max baud rate: 115200 Kbps • I2C Max clock rate: 400 Khz • USB 2.0 Host Max bitrate: 480Mbps • ADC Max Sampling rate: 48 Khz Data width: 16bits • DAC Max Sampling rate: 48 Khz Data width: 16bits • Display Controller 24-bit bus Maximum speed 270 Mhz	• V(OH): 3V(max), 2.4V(min) • V(OL): 0.8V(max), 0V(min) • V(IH): 3V(max), 1.8V(min) • V(IL): 1.2V(max), 0V(min)

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<pre> graph LR ID[Input Device] --> CPU subgraph CPU [Central Processing Unit] CU[Control Unit] ALU[Arithmetic/Logic Unit] end CPU <--> MU[Memory Unit] CPU --> OD[Output Device] </pre> Maximum speed 270 Mhz		

#5.1: Input/Output Interface

- 1) With reference to the computer system diagram (Figure 1) in the case study notes,
- (a) What are the type (input, output or bi-directional) of processor pins you would connect each of the module's data signal to?

Table 1a

No.	Module	Type (Input, Output or Bidirectional)
i.	Buttons	Input
ii.	Wifi	Bi-Directional
iii.	Touch Screen Controller Data	Input/Bi-Directional**
iv.	System Memory	Bi-Directional
v.	Display	Output

#5.1: Input/Output Interface

- (b) Given the following interface requirement, select the appropriate module from the device information list in the case study notes to connect to the processor.

Table 1b

No.	Module	Interface Type
i.	Wifi	SPI
ii.	Touch Screen Controller	UART
iii.	System Memory	Parallel Bus

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3) Wifi Module

(a) Part Number: WIFI0001N

Overview	USB	Electrical Interface
- Max wireless data rate 100Mbps - USB interface	Supports up to USB2.0, 480Mbps.	- V(OH): 5V(max), 4.2V(min) - V(OL): 1.3V(max), 0V(min) - V(IH): 5V(max), 3.6V(min) - V(IL): 1.8V(max), 0V(min)

(b) Part Number: WIFI0010AC

Overview	SPI	Electrical Interface
- Max wireless data rate 100Mbps - SPI interface	- Max Clock rate: 50Mhz - Full duplex operation	- V(OH): 3V(max), 2.4V(min) - V(OL): 0.8V(max), 0V(min) - V(IH): 3V(max), 1.8V(min) - V(IL): 1.2V(max), 0V(min)

(c) Part Number: WIFI0015AC

Overview	SPI	Electrical Interface
- Max wireless data rate 100Mbps - SPI interface	- Max Clock rate: 50Mhz - Full duplex operation	- V(OH): 5V(max), 4.2V(min) - V(OL): 1.3V(max), 0V(min) - V(IH): 5V(max), 3.6V(min) - V(IL): 1.8V(max), 0V(min)

#5.1: Input/Output Interface

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3) Wifi Module

(a) Part Number: WIFI0001N

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- Max wireless data rate 100Mbps - USB interface	Supports up to USB2.0, 480Mbps.	- V(OH): 5V(max), 4.2V(min) - V(OL): 1.3V(max), 0V(min) - V(IH): 5V(max), 3.6V(min) - V(IL): 1.8V(max), 0V(min)

(b) Part Number: WIFI0010AC

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- Max wireless data rate 100Mbps - SPI interface	- Max Clock rate: 50Mhz - Full duplex operation	- V(OH): 3V(max), 2.4V(min) - V(OL): 0.8V(max), 0V(min) - V(IH): 3V(max), 1.8V(min) - V(IL): 1.2V(max), 0V(min)

(c) Part Number: WIFI0015AC

- i. Wifi.
 - a. WIFI0010AC
 - b. 3V electrical interface. VOH, VOL, VIH and VIL compatible with Processor SPI interface.
- ii. Touch Screen Controller
 - a. TS002UART
 - b. 3V electrical interface. VOH, VOL, VIH and VIL compatible with Processor UART interface.
- iii. System Memory
 - a. DRAM0002-16M16, SRAM0002-2M16, NOR0001-1M. Actual choice depends on other requirement such as memory speed (read/write), cost etc.
 - b. 3V electrical interface. VOH, VOL, VIH and VIL compatible with Processor Parallel Bus interface.
 - c. 16 bit data bus compatible with Processor 16 bit parallel bus interface. We can also use 2X 8-bit memory if required.

#5.1: Input/Output Interface

(c) For each of the interface shown in **Table 1b**, is the data transfer serial or parallel?

Table 1b

No.	Module	Interface Type
i.	Wifi	SPI
ii.	Touch Screen Controller	UART
iii.	System Memory	Parallel Bus

Wifi. SPI Bus => 1-bit data in each direction (MOSI, MISO) => Serial Bus. Full duplex.

Touch Screen Controller. UART interface => 1-bit data in each direction (TX, RX) => Serial Bus. Full Duplex.

System Memory. Parallel Bus => multiple data bits transferred simultaneously => Parallel Bus.

#5.1: Input/Output Interface

(c) For each of the interface shown in **Table 1b**, is the data transfer serial or parallel?

(d) Describe the difference between synchronous and asynchronous interface. Are the interfaces shown in **Table 1b** asynchronous or synchronous? Explain.

- i. Wifi – SPI interface. This interface has a common clock line (SCLK) to synchronise the data transfer. => synchronous
- ii. Touch Screen Controller – UART interface. There is no common clock line between transmitter and receiver, all sync information are embedded in the data line signals. => Asynchronous.
- iii. System Memory. Parallel Bus Interface. A Strobe signal is used to tell the receiver when to latch in the data on the data bus. => Synchronous.

#5.1: Input/Output Interface

(e) With reference to the SPI wifi module chosen in 1(b) above, what is the maximum speed that data can be transferred between the wifi module and the processor?

(b) Part Number: WIFI0010AC

i. Overview

- 802.11a/b/g/n/ac support
- Max wireless data rate 100Mbps
- 20x20mm dimension
- SPI interface

ii. Electrical Interface

- V(OH): 3V(max), 2.4V(min)
- V(OL): 0.8V(max), 0V(min)
- V(IH): 3V(max), 1.8V(min)
- V(IL): 1.2V(max), 0V(min)

iii. SPI

- Max Clock rate: 50Mhz
- Full duplex operation

Part Number: Cx1006-200M16

i. Overview

- Von-Neumann Architecture
- 12 Data Registers (32-bit)
- 256 Kbyte on-chip SRAM, 512 Kbyte c EEPROM
- 4 Kbyte Cache (Data and Instruction)
- 400 Mhz CPU clock
- 200 Mhz External Parallel Bus interface
- DMA Controller
- Memory Management Unit

ii. Peripheral

- SPI (1x)
 - a. Maximum clock speed: 50Mhz
- UART (2x)

#5.1: Input/Output Interface

(e) With reference to the SPI wifi module chosen in 1(b) above, what is the maximum speed that data can be transferred between the wifi module and the processor?

(b) Part Number: WIFI0010AC

i. Overview

- 802.11a/b/g/n/ac support
- Max
- 20x
- SPI

ii. Electrical I

- V(C
- V(C
- V(IH): 3V(max), 1.8V(min)
- V(IL): 1.2V(max), 0V(min)

iii. SPI

- Max Clock rate: 50Mhz
- Full duplex operation

Part Number: Cx1006-200M16

i. Overview

- Von-Neumann Architecture

(32-bit)
SRAM, 512 Kbyte c
Data and Instruction)
ck
Parallel Bus interface

Wifi module (WIFI0010AC)

- Max SPI Clock on Wifi Module = 50Mhz

- Max Processor SPI clock = 50Mhz

- => 50Mbps transfer rate (max).

- DMA Controller
- Memory Management Unit

ii. Peripheral

- SPI (1x)
 - a. Maximum clock speed: 50Mhz
- UART (2x)

#5.1: Input/Output Interface

(f) With reference to the device information of the system memory you have chosen in 1(b) above, what is the maximum data transfer rate achievable between the system memory and the processor? Assuming that data on the parallel bus gets transferred on each rising edge of the data strobe.

DRAM (DRAM0002-16M16)

- Max data strobe rate supported by DRAM = 100Mhz
- Max Processor Parallel bus strobe rate = 200Mhz
 - ⇒ Max supported data strobe rate on this link = 100Mhz
- 16-bit interface ⇒ 16 bits data transferred with each strobe
 - ⇒ $100\text{M} * 16 = 1600\text{Mbps}$.

NOR Flash (NOR0001-1M)

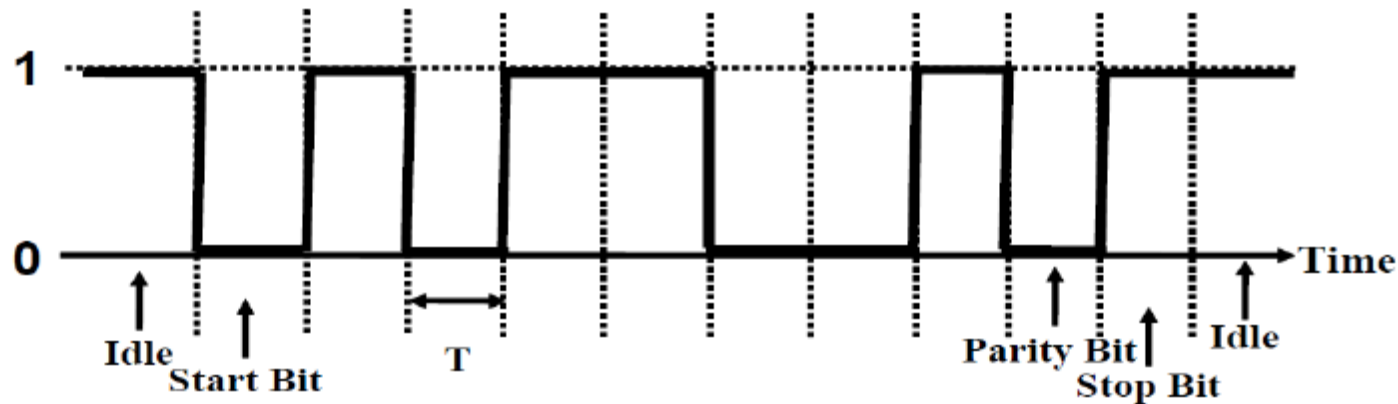
- Max data strobe rate supported by DRAM = 10Mhz
- Max Processor Parallel bus strobe rate = 200Mhz
 - ⇒ Max supported data strobe rate on this link = 10Mhz
- 16-bit interface ⇒ 16 bits data transferred with each strobe
 - ⇒ $10\text{M} * 16 = 160\text{Mbps}$.

SRAM (SRAM0002-2M16)

- Max data strobe rate supported by SRAM = 200Mhz
- Max Processor Parallel bus strobe rate = 200Mhz
 - ⇒ Max supported data strobe rate on this link = 200Mhz
- 16-bit interface ⇒ 16 bits data transferred with each strobe
 - ⇒ $200\text{M} * 16 = 3200\text{Mbps}$.

#5.2: Data Transfer

2) Fig. 2 shows the logical waveform of an asynchronous data frame.



(a) Specify the 7-bit ASCII character that is transmitted (LSB is transmitted first).

7-bits: $1001101_2 \rightarrow 0x4D$

ASCII character $\rightarrow M$

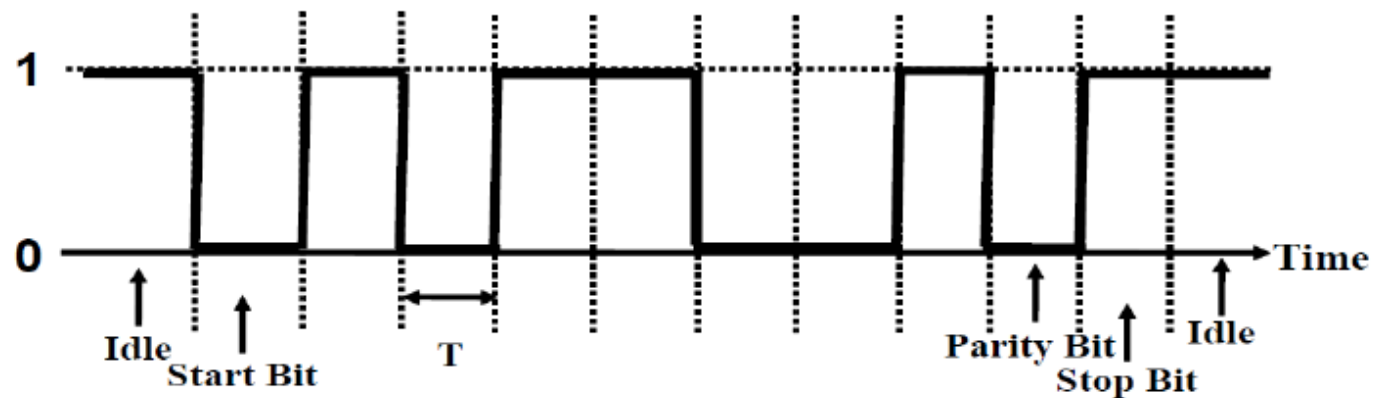
(b) Assume there are no errors in the transmitted waveform, state whether even or odd parity is being used.

Even parity. Data+Parity bits $\rightarrow$ even number of '1's.

(c) Assume two of the bits received in the character are erroneous; can the receiver detect the error? Can you extend the example to determine the limitations of parity checking?
No, the parity of the data sequence remains unchanged if two bits are erroneous. Hence, the parity check will fail. In general, if there is an even number of errors bits, parity checking fails.

#5.2: Data Transfer

2) Fig. 2 shows the logical waveform of an asynchronous data frame.



(d) If 7-bit ASCII characters are transmitted continuously in the format shown (with no idle periods between the frames) at a baud rate of 9600, calculate the following:

- The value of **T** in Fig 2.
- The data transfer rate of this serial interface in characters per second (cps).

(i) $T = 1/9600 = 0.104 \text{ ms}$.

(ii) 1 packet consists of Start, Data, Parity and Stop Bits.

9600 bits per second / 10 bits (1 packet) = 960 characters per second.

(e) Re-compute the data transfer rate (cps) if the transmission does not use any parity bit. Compare the results with those obtained in Q3d(ii), and state your observation(s).

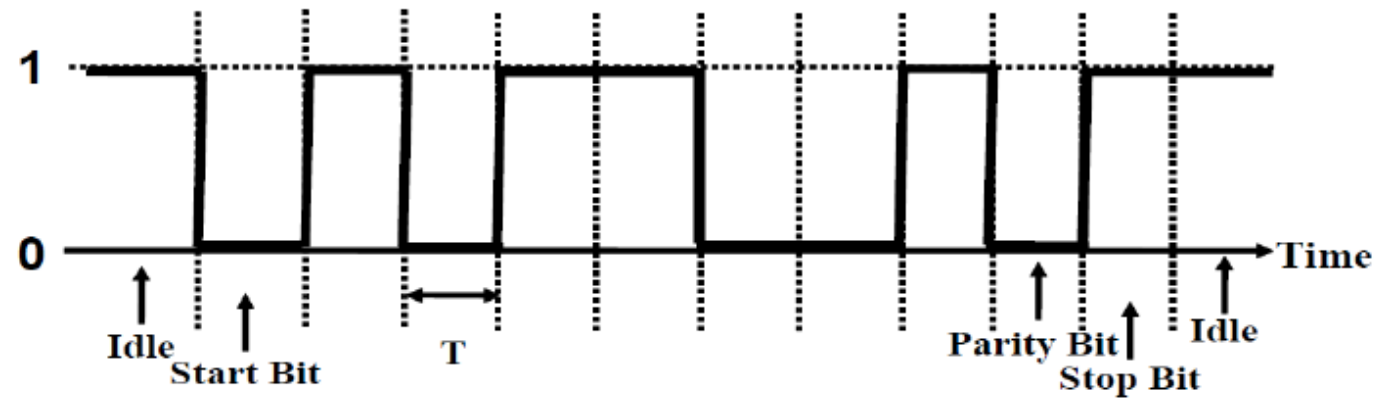
9600 bits per second / 9 bits (1 packet) = 1066 characters per second.

Throughput is higher, but the integrity of the data packet cannot be ascertained.

Which option to choose will depend on the signal line condition (noisy?) between the transmitter and receiver.

#5.2: Data Transfer

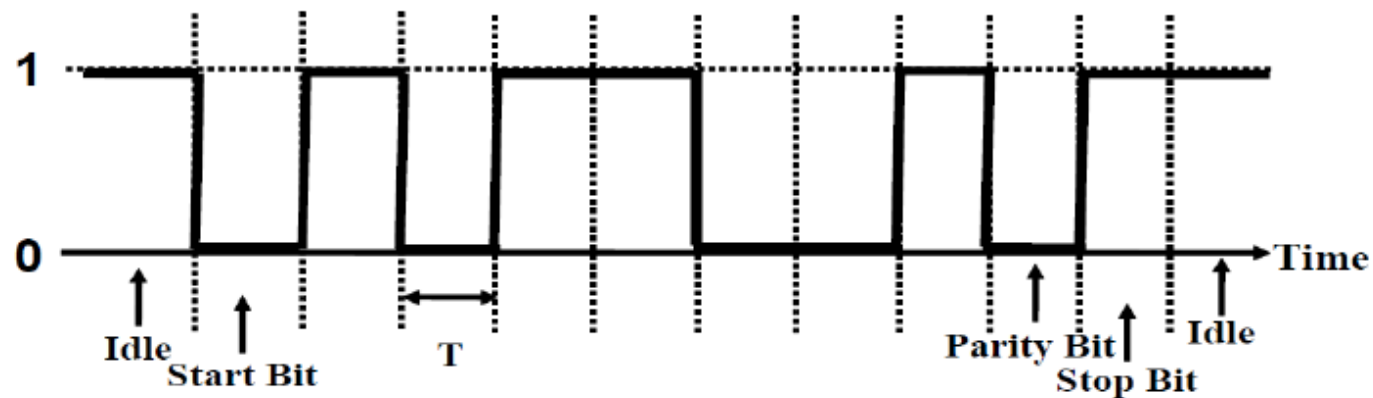
2) Fig. 2 shows the logical waveform of an asynchronous data frame.



- (f) Assume the baud rate of the transmitter is 4800, but the baud rate of the receiver is configured as 9600. Based on Fig. 2, determine, if any, the ASCII character(s) that will be received.

#5.2: Data Transfer

2) Fig. 2 shows the logical waveform of an asynchronous data frame.



(f) Assume the configured will be received

4800 baud rate: 7-bits: $1001101_2 \rightarrow 0x4D$

Original Frame											
S	D (lsb to msb)				P	E					
0	1011001				0	1					
Oversampled Frames (2x)											
Frame #1						Frame #2					
S	D (lsb to msb)				P	E	S	D (lsb to msb)		P	E
0	0110011				1	1	0	0001100		1	1

Frame #1: $0x66 \rightarrow$ lowercase letter 'f'

Frame #2: $0x18 \rightarrow$ special character 'CAN'

Parity errors will be flagged for both frames (odd numbers of '1's received).

Optional questions

- 3) There has been a trend of 'serialization' of interface bus standard. E.g. USB replacing Parallel Port Interface, SATA replacing IDE in HDD.
- (a) What are the advantages that Serial Bus has over Parallel Bus interface that enticed industry player to move in this direction?
 - (b) What are the scenarios in which Parallel bus will still be preferred over Serial bus?
What are the design considerations that need to be put in place in such cases?
- (a) Serial Bus has **less** data lines compared to Parallel Bus.
- > Less issue with signal skew and less probability of being influenced by crosstalk.
 - > Less bulky so easier to route PCB trace and cabling (SATA vs IDE Cable).
 - > Even though data is transferred one bit at a time, modern serial standards are able to achieve relatively high data transfer speed (USB3.0 raw speed is up to 5Gbps). This is achieved through good design and testing conformance/certifications.

Optional questions

- 3) There has been a trend of ‘serialization’ of interface bus standard. E.g. USB replacing Parallel Port Interface, SATA replacing IDE in HDD.
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 - (b) What are the scenarios in which Parallel bus will still be preferred over Serial bus?
What are the design considerations that need to be put in place in such cases?
- (b) Parallel bus is still commonly used when sustained high speed is needed. E.g. interfacing to DDR SDRAM (DDR3 SDRAM data rate scales up to beyond 20GByte/s). But designing these high speed parallel buses require deep knowledge of electrical circuit behaviour, so as to reduce the effect of signal skew and crosstalk. And typically, the length of the parallel bus is very short (order of cm rather than metre), with lots of consideration on grounding and shielding. High speed circuit design is almost an art.

Optional questions

- 4) In Q3(f) above, a wrong sampling result is observed when a wrong baud rate is used. This can actually be used to implement auto baud rate detection. Briefly describe how this can be done.

A Carriage Return has the ASCII value of 0x0D and would be sampled as different value with different baud rate assumption. If the receiving device knows that the transmitting device sends a 0x0D, then it'll be able to derive the baud rate based on the value sampled. Below is an example for the case when receiving device is sampling at 115200 Baud. If the transmitting device is sending 0x0D at 38400 baud, the data received will be 0x1C. Note that other Ascii characters can also be used as long as that value yields different result when sampled under different baud rate.

Optional questions

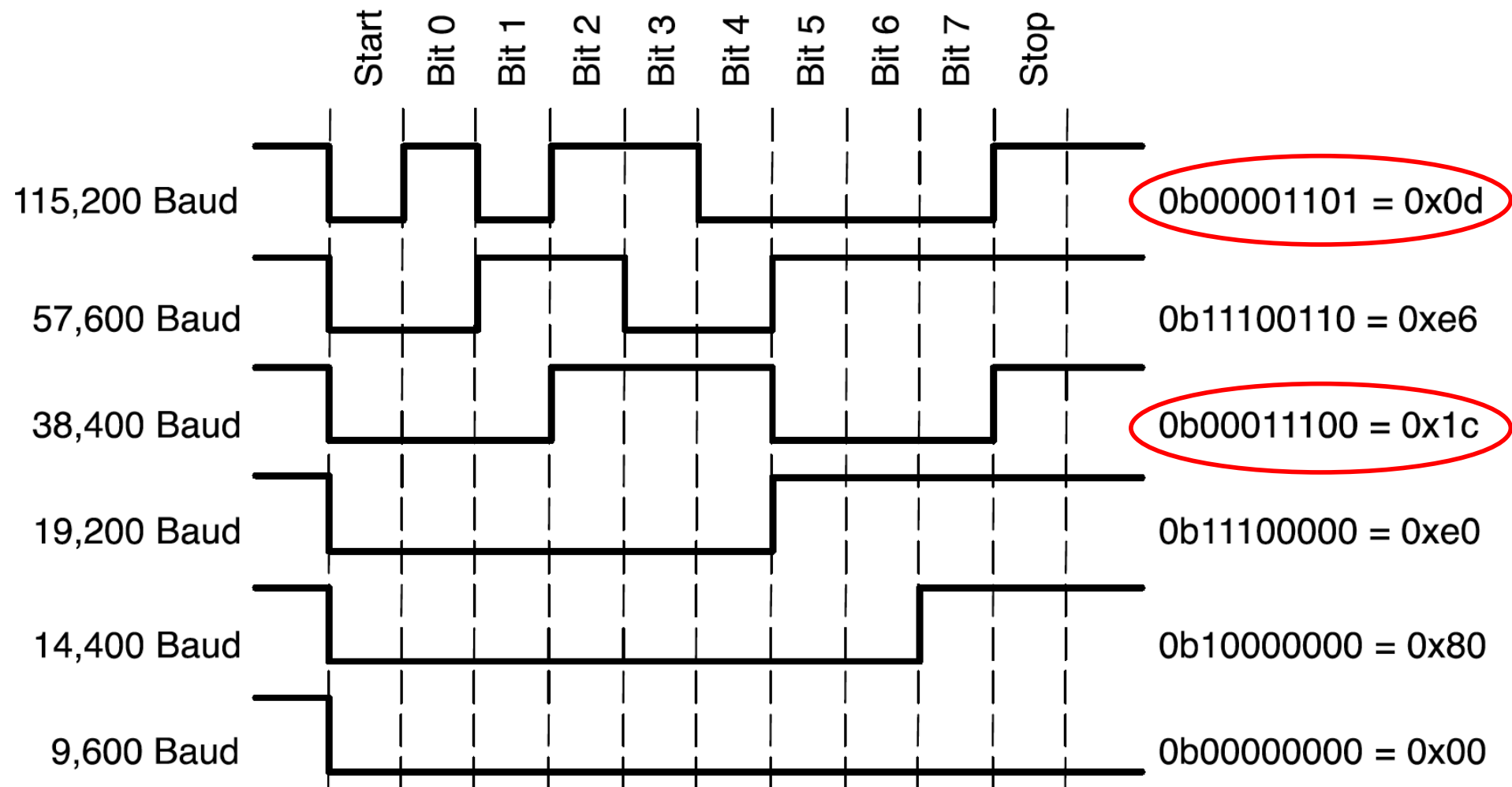


Figure 3. Bit Patterns for Baud Rates Between 115K and 9,600

Optional questions

If data sampled is 0x00. That means baud rate is 9600 or below. In this case, transmitter baud rate has to be adjusted to 9600 and resample the UART data. Below is the chart showing the different values that will be sampled by virtue of the transmitting baud rate.

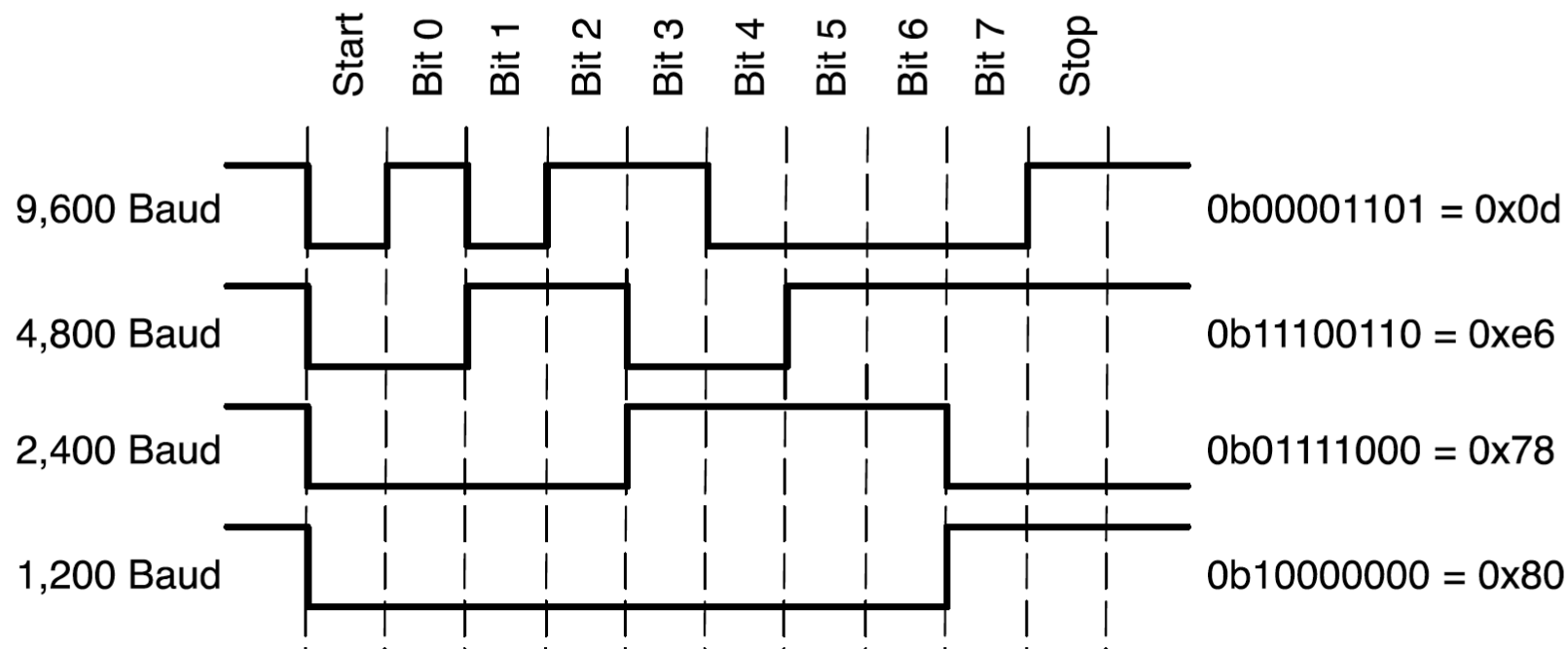


Figure 4. Bit Patterns for Baud Rates Between 9,600 and 1,200